Variational Quantum Eigensolver

Assessment of the VQE algorithm

QUESTION PAPER

Time allowed:	3 hours	**Maximum marks:**	240
Questions:	80 (4 sections × 20)	**Answer key:**	separate Answer Book

Name of the candidate: _____ **Roll No.:** _____

General instructions

1. This question paper has **four sections** of **20 questions** each: **A** Mathematical and Quantum Chemistry Foundations, **B** Quantum Computing Foundations, **C** VQE Fundamentals, **D** Advanced VQE and Implementation.
2. Every question is of the **single-correct** type: four options, exactly one correct. Question formats used: single-correct MCQ, matching lists (List-I/List-II), Assertion–Reason, Statement I/II and paragraph-based questions.
3. For every question choose **ONE** option, (A), (B), (C) or (D), and record it against the question number in the answer script provided by the invigilator.
4. Calculators are **not** allowed. Rough work may be done on separate sheets.
5. Unless a question says otherwise, use the **conventions** printed below; they apply to every section.
6. The detailed solutions are in the separate *Answer Book*; do not consult it before finishing the test.

Marking scheme (every question)

Each question has **four** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

Full marks:	+3 if ONLY the correct option is chosen;
Zero marks:	0 if none of the options is chosen (question unanswered);
Negative marks:	−1 in all other cases (wrong option, or more than one option chosen).

Conventions and notation used throughout the paper

- n qubits; a Pauli string is a word over $\{I, X, Y, Z\}$ whose character k acts on qubit k; basis states $|q_0 q_1 \dots q_{n-1}\rangle$ are written **qubit 0 first**. $H = \sum_k c_k P_k$ with real c_k; E_0 = smallest eigenvalue; $E(\boldsymbol{\theta}) = \langle\psi(\boldsymbol{\theta})|H|\psi(\boldsymbol{\theta})\rangle$.
- $R_Y(\theta) = e^{-i\theta Y/2}$, $R_X(\theta) = e^{-i\theta X/2}$, $R_Z(\theta) = e^{-i\theta Z/2}$, $S = \mathrm{diag}(1, i)$; CUDA-Q's `exp_pauli(theta, q, word)` applies $e^{+i\theta P}$. Two-qubit matrices use the order $|00\rangle, |01\rangle, |10\rangle, |11\rangle$.
- **Jordan–Wigner** (qubit $|1\rangle$ = occupied): $a_j^\dagger = Z_0 \cdots Z_{j-1} \frac{1}{2}(X_j - iY_j)$, $n_j = \frac{1}{2}(I - Z_j)$.
- **The notebook** (`VQE.ipynb`): default Ising chain $H = -J\sum_i Z_i Z_{i+1} - h\sum_i X_i$ (open, $N = 15$, $J = 1$, $h = 0.5$, 29 terms); hardware-efficient ansatz with $L = 2$ layers of [R_Y on all qubits, CNOT ladder] plus a final R_Y layer; COBYLA, 300 evaluations; exact energies via `cudaq.observe`. H_2 example: 4 qubits, 15 Pauli terms (STO-3G, Jordan–Wigner, spin-orbitals ordered $\sigma_g\alpha, \sigma_g\beta, \sigma_u\alpha, \sigma_u\beta$; coefficients correspond to $R = 0.740$ Å), Hartree–Fock $|1100\rangle$, UCC state $e^{i\theta\, X_0X_1X_2Y_3}|1100\rangle$, FCI energy -1.137284 Ha. A *previously recorded run* means the outputs stored in the notebook (CUDA-Q 0.16.0, CPU target `qpp-cpu`).
- Energies of molecules in Hartree (Ha); 1 Ha = 627.51 kcal/mol; chemical accuracy 1.6 mHa. Numerical answers are rounded.

SECTION A · Mathematical and Quantum Chemistry Foundations Q.1–20

Syllabus: eigenvalue problems, Rayleigh quotients, Hermitian matrices, the Schroedinger equation, Born–Oppenheimer, Slater determinants, second quantisation, Pauli decomposition, energy units and correlation energy.
Formats: Single-correct MCQ (Q.1–20); Matching lists (Q.3–11); Assertion–Reason (Q.7–18); Statement I / II (Q.8); Paragraph based (Q.14–15). Each question: +3 / 0 / −1.

Q.1 The ground-state energy (smallest eigenvalue) of $H = \begin{pmatrix} 1 & 2 \\ 2 & -2 \end{pmatrix}$ and a normalised eigenvector for it are

(A) -2 and $(0,1)^{\mathsf{T}}$
(B) 2 and $\frac{1}{\sqrt{5}}(2,1)^{\mathsf{T}}$
(C) -3 and $\frac{1}{\sqrt{5}}(1,-2)^{\mathsf{T}}$
(D) -3 and $\frac{1}{\sqrt{5}}(2,1)^{\mathsf{T}}$

Q.2 For the same matrix $H = \begin{pmatrix} 1 & 2 \\ 2 & -2 \end{pmatrix}$, the Rayleigh quotient $R(v) = \dfrac{v^{\mathsf{T}} H v}{v^{\mathsf{T}} v}$ of the trial vector $v = (1,-1)^{\mathsf{T}}$ is

(A) -5, which lies below E_0
(B) -3, exactly E_0
(C) -2.5, which lies above $E_0 = -3$
(D) 0.5

Q.3 For the real symmetric matrix $M = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 5 \end{pmatrix}$, match each quantity (List-I) with its value (List-II).

List-I		List-II	
(P)	$\operatorname{tr} M$	**(1)**	15
(Q)	$\det M$	**(2)**	1
(R)	smallest eigenvalue of M	**(3)**	9
(S)	largest eigenvalue of M	**(4)**	5

(A) P→3, Q→1, R→4, S→2
(B) P→3, Q→1, R→2, S→4
(C) P→1, Q→3, R→2, S→4
(D) P→3, Q→4, R→2, S→1

Q.4 An energy eigenstate $|\psi_E\rangle$ of a time-independent Hamiltonian ($H|\psi_E\rangle = E|\psi_E\rangle$) evolves under the time-dependent Schrödinger equation $i\hbar\,\partial_t|\psi(t)\rangle = H|\psi(t)\rangle$. Which statement is correct?

(A) $|\psi(t)\rangle = e^{-Et/\hbar}|\psi_E\rangle$: the state decays in time
(B) The state is constant only if $E = 0$; otherwise it moves to other eigenstates
(C) $|\psi(t)\rangle = e^{-iEt/\hbar}|\psi_E\rangle$: only a global phase changes, so all measurement probabilities are constant in time
(D) The probabilities oscillate with angular frequency $E/\hbar$

Q.5 Using 1 Hartree = 627.51 kcal/mol, an energy error of 0.005 Hartree corresponds to about ... and is ... chemically accurate (chemical accuracy: 1 kcal/mol ≈ 1.6 mHa).

(A) 3.14 kcal/mol; not **(B)** 3.14 kcal/mol; still
(C) 0.31 kcal/mol; still **(D)** 125.5 kcal/mol; not

Q.6 In the Born–Oppenheimer approximation the nuclei are held fixed and only the electrons are treated quantum mechanically. In the notebook's 4-qubit H_2 Hamiltonian, the nuclear–nuclear repulsion energy appears

(A) nowhere, because nuclear repulsion is neglected in quantum chemistry
(B) spread over the four `XXYY`-type exchange terms
(C) as the coefficient of the `ZZII` term
(D) inside the coefficient of the identity string `IIII`, as a constant energy shift

Q.7 **Assertion (A):** For every normalised state $|\psi\rangle$, $\langle\psi|H|\psi\rangle \geq E_0$, where E_0 is the smallest eigenvalue of the Hermitian operator H.
Reason (R): Expanding $|\psi\rangle = \sum_k c_k |E_k\rangle$ in the orthonormal eigenbasis of H gives $\langle\psi|H|\psi\rangle = \sum_k |c_k|^2 E_k$ with $\sum_k |c_k|^2 = 1$.

(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A
(B) Both A and R are true, but R is *not* the correct explanation of A
(C) A is true, but R is false
(D) A is false, but R is true

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Q.8 **Statement I:** A real symmetric matrix can have complex (non-real) eigenvalues.
Statement II: Eigenvectors of a Hermitian matrix that belong to different eigenvalues are orthogonal.

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true, but Statement II is false
(D) Statement I is false, but Statement II is true

Q.9 H_2 in the STO-3G basis has 2 spatial orbitals, i.e. 4 spin-orbitals, and 2 electrons. The number of 2-electron Slater determinants, and the number of them with one spin-up and one spin-down electron ($S_z = 0$), are

(A) 6 and 2 **(B)** 6 and 4 **(C)** 16 and 4 **(D)** 4 and 2

Q.10 LiH in the minimal STO-3G basis has one $1s$ orbital on H and the orbitals $1s, 2s, 2p_x, 2p_y, 2p_z$ on Li. With the Jordan–Wigner mapping (one qubit per spin-orbital, no reductions), the number of qubits and the number of electrons are

(A) 24 qubits, 4 electrons **(B)** 12 qubits, 3 electrons
(C) 12 qubits, 4 electrons **(D)** 6 qubits, 4 electrons

Q.11 Fermionic creation and annihilation operators $a_p^\dagger$, a_p satisfy the canonical anticommutation relations. Match each expression (List-I) with its value (List-II); $|\text{vac}\rangle$ is the empty (vacuum) state.

List-I		**List-II**	
(P)	$\{a_p, a_q^\dagger\} = a_p a_q^\dagger + a_q^\dagger a_p$	**(1)**	$\{0, 1\}$
(Q)	$a_p^\dagger a_p^\dagger$	**(2)**	$\delta_{pq}\, I$
(R)	spectrum of $n_p = a_p^\dagger a_p$	**(3)**	0 (the zero vector)
(S)	$a_p \,\lvert\text{vac}\rangle$	**(4)**	0 (the zero operator)

(A) P→2, Q→4, R→1, S→3 **(B)** P→2, Q→3, R→1, S→4
(C) P→2, Q→4, R→3, S→1 **(D)** P→4, Q→2, R→1, S→3

Q.12 In second quantisation $H = \sum_{pq} h_{pq} a_p^\dagger a_q + \frac{1}{2}\sum_{pqrs} h_{pqrs} a_p^\dagger a_q^\dagger a_r a_s$ (+ constant). For N spin-orbitals the number of terms in this Hamiltonian grows as

(A) $O(N)$, one term per orbital
(B) $O(N^4)$, from the two-electron integrals h_{pqrs}
(C) $O(N^2)$, from the one-electron integrals only
(D) $O(2^N)$, like the dimension of the Fock space

Q.13 For a Pauli operator P ($P^2 = I$), $e^{i\theta P} = \cos\theta\, I + i \sin\theta\, P$. Hence $e^{i\pi X/4}$ equals

(A) $\begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$ **(B)** $\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ **(C)** $\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}$ **(D)** $\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -i \\ -i & 1 \end{pmatrix}$

Paragraph for Questions 14 and 15

A single qubit has the Hamiltonian $H = -Z + \frac{1}{2}X = \begin{pmatrix} -1 & \frac{1}{2} \\ \frac{1}{2} & 1 \end{pmatrix}$. A one-parameter trial state is prepared as $|\psi(\theta)\rangle = R_Y(\theta)\,|0\rangle = \cos\frac{\theta}{2}\,|0\rangle + \sin\frac{\theta}{2}\,|1\rangle$, and its energy $E(\theta) = \langle\psi(\theta)|H|\psi(\theta)\rangle$ is minimised over $\theta \in (-\pi, \pi]$.

Q.14 The energy function $E(\theta)$ and the exact ground-state energy E_0 are

(A) $E(\theta) = -\cos\theta + \frac{1}{2}\sin\theta$ and $E_0 = -1$
(B) $E(\theta) = -\cos\frac{\theta}{2} + \frac{1}{2}\sin\frac{\theta}{2}$ and $E_0 = -\frac{\sqrt{5}}{2}$
(C) $E(\theta) = -\cos\theta + \sin\theta$ and $E_0 = -\sqrt{2}$
(D) $E(\theta) = -\cos\theta + \frac{1}{2}\sin\theta$ and $E_0 = -\frac{\sqrt{5}}{2} \approx -1.118$

Q.15 The optimal angle θ^*, and the energy error $E(0) - E_0$ of the starting point $\theta = 0$ (the state $|0\rangle$), are

(A) $\theta^* = -\arctan\frac{1}{2} \approx -0.464$ rad; error 0
(B) $\theta^* = -\arctan 2 \approx -1.107$ rad; error ≈ 0.118
(C) $\theta^* = +\arctan\frac{1}{2} \approx 0.464$ rad; error ≈ 0.118
(D) $\theta^* = -\arctan\frac{1}{2} \approx -0.464$ rad; error ≈ 0.118

Q.16 The Hermitian matrix $M = \begin{pmatrix} 3 & 1-2i \\ 1+2i & -1 \end{pmatrix}$ written as $aI + bX + cY + dZ$ is

(A) $I + X + 2Y + 2Z$ **(B)** $3I + X + 2Y - Z$ **(C)** $I + X - 2Y + 2Z$ **(D)** $2I + X + 2Y + Z$

Q.17 The diagonal two-qubit Hamiltonian $H = \mathrm{diag}(1, 2, 3, 4)$ (basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$) written as a sum of Pauli strings is

(A) $2.5\, II - 0.5\, ZI - 1.0\, IZ$

(B) $2.5\, II - 1.0\, ZI - 0.5\, IZ + 0.5\, ZZ$

(C) $2.5\, II + 1.0\, ZI + 0.5\, IZ$

(D) $2.5\, II - 1.0\, ZI - 0.5\, IZ$

Q.18 **Assertion (A):** For H_2 at large bond length, the restricted Hartree–Fock (single closed-shell determinant) energy lies far above the exact energy.

Reason (R): The nuclear–nuclear repulsion $1/R$ tends to zero as $R \to \infty$.

(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A

(B) Both A and R are true, but R is *not* the correct explanation of A

(C) A is true, but R is false

(D) A is false, but R is true

Q.19 For the notebook's H_2 Hamiltonian, the Hartree–Fock energy is $E_{\mathrm{HF}} = \langle 1100|H|1100\rangle = -1.116759$ Ha and the FCI energy is -1.137284 Ha. The correlation energy $E_{\mathrm{corr}} = E_{\mathrm{FCI}} - E_{\mathrm{HF}}$ is about

(A) -20.5 mHa **(B)** $+20.5$ mHa **(C)** -2.05 mHa **(D)** -1137 mHa

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Q.20 A trial state of H_2 has 90 % weight on the ground state ($E_0 = -1.13728$ Ha) and 10 % on another eigenstate of the 4-qubit Hamiltonian with energy $E_1 = -0.53821$ Ha. Its energy expectation and its error above E_0 are

(A) -1.1971 Ha and -59.9 mHa

(B) -1.0774 Ha and 59.9 mHa

(C) -0.8378 Ha and 299.5 mHa

(D) -1.1373 Ha and 0 mHa

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SECTION B · Quantum Computing Foundations Q.21–40

Syllabus: qubits and the Bloch sphere, rotation gates, Pauli algebra and tensor products, measurement bases, CNOT and entanglement, qubit ordering, `exp_pauli`, commuting observables, memory and variance.
Formats: Single-correct MCQ (Q.21–40); Matching lists (Q.23–24); Assertion–Reason (Q.32); Statement I / II (Q.28–39); Paragraph based (Q.36–37). Each question: +3 / 0 / −1.

Q.21 The state $R_Y(\pi/3)\,|0\rangle$ is measured in the computational basis. The probability of the outcome 1 is

(A) $\frac{3}{4}$ **(B)** $\frac{\sqrt{3}}{2}$ **(C)** $\frac{1}{2}$ **(D)** $\frac{1}{4}$

Q.22 The two-qubit operator $(X \otimes Y)(Y \otimes X)$ equals

(A) $Z \otimes Z$ **(B)** $Y \otimes Y$ **(C)** $-Z \otimes Z$ **(D)** $i\,Z \otimes Z$

Q.23 Match each single-qubit state (List-I) with its Bloch vector $(\langle X\rangle, \langle Y\rangle, \langle Z\rangle)$ (List-II).

List-I		List-II	
(P)	$\|0\rangle$	**(1)**	$(0, 1, 0)$
(Q)	$\|-\rangle = \frac{1}{\sqrt{2}}(\|0\rangle - \|1\rangle)$	**(2)**	$(\frac{\sqrt{3}}{2}, 0, \frac{1}{2})$
(R)	$\frac{1}{\sqrt{2}}(\|0\rangle + i\,\|1\rangle)$	**(3)**	$(0, 0, 1)$
(S)	$R_Y(\pi/3)\,\|0\rangle$	**(4)**	$(-1, 0, 0)$

(A) P→3, Q→4, R→2, S→1 **(B)** P→3, Q→1, R→4, S→2
(C) P→4, Q→3, R→1, S→2 **(D)** P→3, Q→4, R→1, S→2

Q.24 To estimate $\langle P\rangle$ from computational-basis measurements, each Pauli string is first rotated into a string of Z's and I's. Match each observable (List-I) with the gates applied before measuring (List-II).

List-I		List-II	
(P)	X	**(1)**	H on qubit 0 only
(Q)	Y	**(2)**	nothing
(R)	Z	**(3)**	H
(S)	$X_0 Z_1$	**(4)**	$S^\dagger$, then H

(A) P→3, Q→2, R→4, S→1 **(B)** P→3, Q→4, R→1, S→2
(C) P→3, Q→4, R→2, S→1 **(D)** P→4, Q→3, R→2, S→1

Q.25 For the Bell state $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$, the expectation value $\langle Y \otimes Y\rangle$ is

(A) $+1$ **(B)** $-\frac{1}{2}$ **(C)** 0 **(D)** -1

Q.26 The Hartree–Fock state of the notebook's H_2 example is $|1100\rangle$, written with qubit 0 first. In the vector returned by `cudaq.get_state`, which lists amplitudes with qubit 0 as the *least* significant bit, this basis state has the index

(A) 12 **(B)** 3 **(C)** 0 **(D)** 6

Q.27 Starting from $|00\rangle$, apply $R_Y(\pi/2)$ to qubit 0 and then a CNOT with control 0 and target 1. The resulting state is

(A) $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ **(B)** $\frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$
(C) $\frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$ **(D)** $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$

Q.28 **Statement I:** Any circuit of gates applied to $|0\ldots0\rangle$ prepares a normalised state, so the ansatz state $|\psi(\boldsymbol{\theta})\rangle$ needs no explicit normalisation.

Statement II: Measuring in the computational basis directly reveals the eigenvalue of an X_i term, without any basis change.

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true, but Statement II is false
(D) Statement I is false, but Statement II is true

Q.29 Consider the four exchange terms of the notebook's H_2 Hamiltonian, `XXYY`, `XYYX`, `YXXY`, `YYXX`. Which statement is correct?

(A) Every pair anticommutes
(B) All six pairs commute, but no pair is qubit-wise commuting
(C) Only `XXYY` and `YYXX` commute
(D) All six pairs are qubit-wise commuting, so one measurement setting suffices

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Q.30 CUDA-Q's `exp_pauli(theta, q, word)` applies $e^{i\theta P}$ for the Pauli string P given by `word`. Applied with the word `"ZZ"` to the basis state $|01\rangle$, it gives

(A) $\cos\theta\,|01\rangle + i\sin\theta\,|10\rangle$ **(B)** $e^{+i\theta}\,|01\rangle$
(C) $e^{-i\theta}\,|01\rangle$ **(D)** $|10\rangle$

Q.31 The four-qubit Pauli string $X_0X_1X_2Y_3$ (word `"XXXY"`) acting on $|1100\rangle$ gives

(A) $i\,|0011\rangle$ **(B)** $|0011\rangle$ **(C)** $-i\,|0011\rangle$ **(D)** $i\,|1100\rangle$

Q.32 **Assertion (A):** For a Hermitian H, the expectation value $\langle\psi|H|\psi\rangle$ is a real number for every state $|\psi\rangle$.

Reason (R): The identity $\langle\psi|H|\psi\rangle = \langle\psi|H|\psi\rangle^*$ holds only for normalised states.

(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A
(B) Both A and R are true, but R is *not* the correct explanation of A
(C) A is true, but R is false
(D) A is false, but R is true

Q.33 A complex fp64 amplitude needs 16 bytes. For LiH in STO-3G with Jordan–Wigner (12 qubits), the memory needed for the state vector and for the density matrix ($2^n \times 2^n$ amplitudes) is

(A) 192 B and 2.3 kB **(B)** 65.5 kB and 131 kB
(C) 268 MB and 65.5 kB **(D)** 65.5 kB and 268 MB

Q.34 Two qubits start in $|00\rangle$; $R_Y(\theta_0)$ and $R_Y(\theta_1)$ are applied to qubits 0 and 1, followed by a CNOT (control 0, target 1). The expectation value $\langle Z_0 Z_1 \rangle$ of the final state is

(A) $\cos(\theta_0 + \theta_1)$ **(B)** $\cos\theta_1$ **(C)** $\cos\theta_0$ **(D)** $\cos\theta_0 \cos\theta_1$

Q.35 Which unitary U satisfies $UYU^\dagger = Z$, so that measuring Z after applying U measures Y?

(A) $U = S$ **(B)** $U = HS^\dagger$ (first $S^\dagger$, then H)
(C) $U = SH$ (first H, then S) **(D)** $U = H$

Paragraph for Questions 36 and 37

A two-qubit transverse-field Ising Hamiltonian $H = -Z_0 Z_1 - \frac{1}{2}(X_0 + X_1)$ (the notebook's `"tfim"` model with $N = 2$, $J = 1$, $h = 0.5$) is treated with a tiny hardware-efficient circuit: starting from $|00\rangle$, apply $R_Y(\theta_0)$ to qubit 0 and $R_Y(\theta_1)$ to qubit 1, then a CNOT with control 0 and target 1. The angles are $\theta_0 = \pi/2$, $\theta_1 = 0$.

Q.36 The energy $\langle\psi|H|\psi\rangle$ of the prepared state is

(A) -1 **(B)** 0 **(C)** $-\sqrt{2} \approx -1.414$ **(D)** -1.5

Q.37 The exact ground-state energy E_0 of this H, and the relative error of the prepared state, are

(A) -1; 0 % **(B)** $-\sqrt{2} \approx -1.414$; 29.3 %
(C) $-\sqrt{2}$; 41.4 % **(D)** -1.5; 33.3 %

Q.38 Every Hermitian $2^n \times 2^n$ matrix is a real linear combination of the n-qubit Pauli strings. For $n = 4$ the number of Pauli strings, and the number actually needed for the notebook's H_2 Hamiltonian, are

(A) 256 and 256 **(B)** 64 and 15 **(C)** 16 and 15 **(D)** 256 and 15

Q.39 **Statement I:** Two Pauli strings either commute or anticommute.
Statement II: A set of mutually commuting Pauli strings can be measured simultaneously after one common basis change.

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true, but Statement II is false
(D) Statement I is false, but Statement II is true

Q.40 For $H = -Z + \frac{1}{2}X$ (one qubit), the energy variance $\mathrm{Var}(H) = \langle H^2 \rangle - \langle H \rangle^2$ in the state $|0\rangle$, and in the exact ground state, are

(A) 0.25 and 1.25 **(B)** 1.25 and 0 **(C)** 0.25 and 0 **(D)** 0 and 0.25

SECTION C · VQE Fundamentals Q.41–60

Syllabus: variational and Rayleigh–Ritz principles, ansatz design, Hartree–Fock reference, expectation values, Hamiltonian decomposition, measurement grouping, classical optimizers, convergence and the notebook's code.
Formats: Single-correct MCQ (Q.41–57); Matching lists (Q.46–58); Assertion–Reason (Q.49); Statement I / II (Q.52); Paragraph based (Q.59–60). Each question: +3 / 0 / −1.

Q.41 A VQE run with *exact* expectation values (no shots, no noise) reports $E(\boldsymbol{\theta}) = -15.10$ for the 15-qubit Ising chain whose exact ground energy is $E_0 = -15.0825$. The most likely explanation is

(A) the optimizer found a state better than the exact ground state

(B) a deeper ansatz always produces energies below the exact value

(C) an error in the Hamiltonian, the reference energy or the state normalisation — a correct calculation cannot go below E_0

(D) shot noise pushed the estimate below E_0

Q.42 In the Rayleigh–Ritz method, H is diagonalised inside an m-dimensional trial subspace, giving Ritz values $\varepsilon_1 \le \cdots \le \varepsilon_m$. Which statement is correct?

(A) Only ε_1 is bounded; the other Ritz values can lie below the exact eigenvalues

(B) The Ritz values are lower bounds to the exact eigenvalues

(C) $\varepsilon_k \ge E_{k-1}$ for every k: each Ritz value is an upper bound to the corresponding exact eigenvalue

(D) The Ritz values are exact whenever $m \ge 2$

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Q.43 The notebook's hardware-efficient ansatz uses L layers of [R_Y on every qubit, CNOT ladder over neighbours] plus a final R_Y layer. For $N = 10$ qubits and $L = 3$, the numbers of parameters, CNOTs and gates are

(A) 40, 30 and 70 **(B)** 30, 27 and 57 **(C)** 40, 9 and 49 **(D)** 40, 27 and 67

Q.44 The UCC kernel starts with `x(qubits[0]); x(qubits[1])`. In the Jordan–Wigner picture with spin-orbitals ordered (orbital 0 α, orbital 0 β, orbital 1 α, orbital 1 β), this prepares

(A) the state with both electrons in the antibonding orbital

(B) an equal superposition of all 2-electron determinants

(C) the Hartree–Fock determinant $|1100\rangle$: both electrons in the lower (bonding) spatial orbital, one with each spin

(D) a triplet state with both electrons spin-up

Q.45 Let $|u\rangle$, $|v\rangle$ be orthonormal, $H_{uu} = a$, $H_{vv} = c$, $H_{uv} = H_{vu} = b$ (real). For the trial state $|\psi(\theta)\rangle = \cos\theta\,|u\rangle - \sin\theta\,|v\rangle$, the energy is

(A) $a\cos^2\theta + c\sin^2\theta - b\sin 2\theta$ **(B)** $a\cos^2\theta - c\sin^2\theta$

(C) $a\cos\theta + c\sin\theta - b$ **(D)** $a\cos^2\theta + c\sin^2\theta + b\sin 2\theta$

Q.46 Match each VQE ingredient (List-I) with the code in `VQE.ipynb` that implements it (List-II).

List-I		List-II	
(P)	Hartree–Fock reference state	**(1)**	`minimize(energy, initial_theta, method="COBYLA", ...)`
(Q)	energy $E(\boldsymbol{\theta}) = \langle\psi(\boldsymbol{\theta})\|H\|\psi(\boldsymbol{\theta})\rangle$	**(2)**	`coefficient * cudaq.SpinOperator.from_word(word)`, summed
(R)	classical parameter update	**(3)**	`x(qubits[0]); x(qubits[1])`
(S)	Hamiltonian as a sum of Pauli strings	**(4)**	`cudaq.observe(kernel, hamiltonian, *args).expectation()`

(A) P→3, Q→4, R→1, S→2 **(B)** P→3, Q→4, R→2, S→1
(C) P→4, Q→3, R→1, S→2 **(D)** P→3, Q→1, R→4, S→2

Q.47 $H = 0.5\,Z_0 - 0.3\,Z_1 + 0.2\,X_0X_1$. A trial state has $\langle Z_0\rangle = 0.6$, $\langle Z_1\rangle = -0.2$ and $\langle X_0X_1\rangle = 0.5$. Its energy is

(A) 0.34 **(B)** 1.3 **(C)** 0.46 **(D)** 0.26

Q.48 Only the identity and the Z-type terms of the notebook's H_2 Hamiltonian contribute to $\langle 1100|H|1100\rangle$. With coefficients $c_{IIII} = -0.097066$, $c_{ZIII} = c_{IZII} = 0.171413$, $c_{IIZI} = c_{IIIZ} = -0.223432$, $c_{ZZII} = 0.168689$, $c_{ZIZI} = c_{IZIZ} = 0.120625$, $c_{ZIIZ} = c_{IZZI} = 0.165928$, $c_{IIZZ} = 0.174413$, the Hartree–Fock energy is

(A) −1.137284 Ha **(B)** −0.097066 Ha **(C)** +0.462618 Ha **(D)** −1.116759 Ha

Q.49 **Assertion (A):** The notebook's one-parameter UCC ansatz can reach the exact (FCI) ground energy of its H_2 Hamiltonian.

Reason (R): The exact ground state of that Hamiltonian lies in span$\{|1100\rangle, |0011\rangle\}$, and $\cos\theta\,|1100\rangle - \sin\theta\,|0011\rangle$ covers every real normalised vector of that span (up to sign).

(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A
(B) Both A and R are true, but R is *not* the correct explanation of A
(C) A is true, but R is false
(D) A is false, but R is true

Q.50 For the default Ising chain ($N = 15$, $J = 1$, $h = 0.5$, $H = -J\sum Z_iZ_{i+1} - h\sum X_i$), the energy of $|00\ldots0\rangle$ — close to where the hardware-efficient ansatz starts, since its initial angles are drawn from $\mathcal{N}(0, 0.1^2)$ — is

(A) −15 **(B)** −21.5 **(C)** −7.5 **(D)** −14

Q.51 Each of the 45 angles of the default hardware-efficient ansatz enters exactly one R_Y gate. The number of circuit evaluations for one exact parameter-shift gradient, and for one forward finite-difference gradient (L-BFGS-B), are

(A) 90 and 90 **(B)** 90 and 46 **(C)** 2 and 46 **(D)** 45 and 46

Q.52 **Statement I:** COBYLA needs no gradient evaluations.

Statement II: The notebook reports the energy of the optimizer's final point `opt.x` as the VQE energy.

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true, but Statement II is false

(D) Statement I is false, but Statement II is true

Q.53 Four VQE runs on the notebook's H_2 Hamiltonian (exact expectation values; FCI = -1.137284 Ha) report the energies below. Which one is a valid, chemically accurate result?

(A) -1.11676 Ha **(B)** -1.13600 Ha **(C)** -1.13500 Ha **(D)** -1.13900 Ha

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Q.54 The notebook's H_2 Hamiltonian has the identity, 10 Z-type terms and 4 terms `XXYY`, `XYYX`, `YXXY`, `YYXX`. With *qubit-wise commuting* grouping, the minimum number of measurement settings is ..., and with general commuting grouping ...

(A) 5; 2 **(B)** 15; 5 **(C)** 2; 1 **(D)** 4; 1

Q.55 In the previously recorded default run, $E_{\mathrm{VQE}} = -15.0533898284$ and the exact $E_0 = -15.0825179883$. The relative error is

(A) 1.93 % **(B)** 0.193 % **(C)** 0.0291 % **(D)** -0.193 %

Q.56 The notebook's hardware-efficient ansatz uses only R_Y rotations and CNOTs, so it produces states with *real* amplitudes. Why is this adequate for its Ising and H_2 Hamiltonians?

(A) Complex amplitudes cannot be measured, so they are never needed

(B) Both Hamiltonians are real symmetric matrices, so a ground state can always be chosen real

(C) Real amplitudes always give lower energies than complex ones

(D) CNOT gates generate all complex phases automatically

Q.57 The recorded optimised state of the Ising chain gives $|000\ldots0\rangle$ with probability 0.70 but $|111\ldots1\rangle$ (not in the top 8) with nearly zero probability, although H is symmetric under flipping all spins. The best explanation is

(A) $|111\ldots1\rangle$ has a higher energy than $|000\ldots0\rangle$

(B) the exact ground state contains only $|000\ldots0\rangle$

(C) the simulator has a bias towards $|0\rangle$

(D) for $h < J$ the two ordered configurations are nearly degenerate; VQE settled on one of them (symmetry breaking), which costs only an energy exponentially small in N

Q.58 Match each optimizer offered by the notebook (List-I) with its working principle (List-II).

List-I		List-II	
(P)	COBYLA	**(1)**	successive line searches along a set of conjugate directions, derivative-free
(Q)	Nelder–Mead	**(2)**	quasi-Newton method with a limited-memory Hessian estimate, using gradients (finite differences here) and box bounds
(R)	Powell	**(3)**	linear models of the objective inside a shrinking trust region, derivative-free
(S)	L-BFGS-B	**(4)**	a simplex of $n+1$ points updated by reflection, expansion and contraction

(A) P→3, Q→4, R→1, S→2 **(B)** P→3, Q→1, R→4, S→2
(C) P→4, Q→3, R→1, S→2 **(D)** P→3, Q→4, R→2, S→1

Paragraph for Questions 59 and 60

In the notebook's H_2 example the UCC state $|\psi(\theta)\rangle = \cos\theta\,|1100\rangle - \sin\theta\,|0011\rangle$ only involves two basis states. Within this two-dimensional space the Hamiltonian is the real 2×2 matrix

$$\begin{pmatrix} H_{11} & H_{12} \\ H_{12} & H_{22} \end{pmatrix} = \begin{pmatrix} -1.116759 & 0.181210 \\ 0.181210 & 0.462618 \end{pmatrix}\ \text{Ha},$$

with $H_{11} = \langle 1100|H|1100\rangle$, $H_{22} = \langle 0011|H|0011\rangle$, $H_{12} = \langle 1100|H|0011\rangle$ (computed from the notebook's 15 Pauli terms), so that $E(\theta) = H_{11}\cos^2\theta + H_{22}\sin^2\theta - H_{12}\sin 2\theta$.

Q.59 The lowest eigenvalue of the 2×2 matrix — the best energy the UCC ansatz can reach — is

(A) -1.116759 Ha **(B)** $+0.483143$ Ha **(C)** -0.654141 Ha **(D)** -1.137284 Ha

Q.60 The optimal angle θ^* and the probability of measuring $|0011\rangle$ in the optimal state are

(A) $\theta^* \approx 0.2295$ rad and 0.0518 **(B)** $\theta^* \approx 0.1128$ rad and 0.0127
(C) $\theta^* \approx -0.1128$ rad and 0.0127 **(D)** $\theta^* \approx 0.1128$ rad and 0.1125

SECTION D · Advanced VQE and Implementation Q.61–80

Syllabus: Jordan–Wigner and Bravyi–Kitaev mappings, UCC, molecular scaling, shots, noise and error mitigation, backends, exact references, benchmarking and interpretation of recorded results.
Formats: Single-correct MCQ (Q.61–78); Matching lists (Q.64–68); Assertion–Reason (Q.71); Statement I / II (Q.74); Paragraph based (Q.79–80). Each question: +3 / 0 / −1.

Q.61 With the Jordan–Wigner transformation $a_j^\dagger = \left(\prod_{k<j} Z_k\right) \frac{1}{2}(X_j - iY_j)$ (qubit $|1\rangle$ = occupied), the creation operator $a_2^\dagger$ on 4 spin-orbitals becomes

(A) $\frac{1}{2}\, Z_0 Z_1 (X_2 + iY_2)$ **(B)** $\frac{1}{2}(X_2 - iY_2)$
(C) $\frac{1}{2}\, Z_3 (X_2 - iY_2)$ **(D)** $\frac{1}{2}\, Z_0 Z_1 (X_2 - iY_2)$

Q.62 Under Jordan–Wigner, the hopping term $a_0^\dagger a_1 + a_1^\dagger a_0$ between neighbouring spin-orbitals 0 and 1 becomes

(A) $X_0 X_1$ **(B)** $\frac{1}{2}(X_0X_1 + Y_0Y_1)$
(C) $\frac{1}{2}(Z_0 + Z_1)$ **(D)** $\frac{1}{2}(X_0X_1 - Y_0Y_1)$

Q.63 The Pauli weight (number of non-identity factors) of the image of $a_7^\dagger$ on 8 spin-orbitals is ... under Jordan–Wigner and ... under Bravyi–Kitaev.

(A) 8; 4 **(B)** 8; 8 **(C)** 1; 4 **(D)** 4; 8

Q.64 Match each fermion-to-qubit encoding (List-I) with its characteristic property (List-II).

List-I		List-II	
(P)	Jordan–Wigner	**(1)**	qubit j stores the parity $n_0 \oplus \cdots \oplus n_j$; total parity and S_z-parity sit on single qubits, so two qubits can be removed
(Q)	Bravyi–Kitaev	**(2)**	symmetries ($\mathbb{Z}_2$ stabilisers commuting with H) are used to remove qubits from any encoding
(R)	parity mapping	**(3)**	qubit j stores the occupation n_j; strings of length $O(N)$
(S)	qubit tapering	**(4)**	qubits store partial sums of occupations on a binary tree; strings of length $O(\log N)$

(A) P→4, Q→3, R→1, S→2 **(B)** P→3, Q→4, R→1, S→2
(C) P→3, Q→1, R→4, S→2 **(D)** P→3, Q→4, R→2, S→1

Q.65 For 4 modes, the Bravyi–Kitaev qubit values are $b = Bn \bmod 2$ with $B = \begin{pmatrix} 1&0&0&0 \\ 1&1&0&0 \\ 0&0&1&0 \\ 1&1&1&1 \end{pmatrix}$ and n the occupation vector. The Hartree–Fock occupation $n = (1, 1, 0, 0)$ is encoded as

(A) $|0100\rangle$ **(B)** $|1100\rangle$ **(C)** $|1101\rangle$ **(D)** $|1000\rangle$

Q.66 Counting excitations in spin-orbitals without spin restrictions, LiH in STO-3G (12 spin-orbitals, 4 electrons) has how many single and double excitations from the Hartree–Fock determinant?

(A) 32 singles and 168 doubles
(B) 32 singles and 1024 doubles
(C) 8 singles and 28 doubles
(D) 48 singles and 168 doubles

Q.67 Under Jordan–Wigner, the anti-Hermitian double-excitation generator $a_3^\dagger a_2^\dagger a_1 a_0 - a_0^\dagger a_1^\dagger a_2 a_3$ becomes a sum of how many Pauli strings, and what does the notebook's UCC kernel use?

(A) 4 strings with coefficients $\pm\frac{1}{2}$; the notebook keeps `XXYY`
(B) 1 string; the notebook uses all of it
(C) 8 strings with coefficients $\pm i/8$; the notebook keeps the single string `XXXY`
(D) 16 strings; the notebook keeps 8

Q.68 Match each practical VQE difficulty (List-I) with a standard remedy (List-II).

List-I		List-II	
(P)	barren plateaus (vanishing gradients for deep random circuits)	**(1)**	error mitigation, e.g. zero-noise extrapolation
(Q)	shot noise in energy estimates	**(2)**	chemically motivated ansatz, local cost functions, careful (e.g. Hartree–Fock) initialisation
(R)	gate errors on hardware	**(3)**	several restarts / good initial parameters
(S)	local minima of $E(\boldsymbol{\theta})$	**(4)**	measurement grouping and more shots for terms with large $\lvert c_k\rvert$

(A) P→4, Q→2, R→1, S→3
(B) P→2, Q→4, R→1, S→3
(C) P→2, Q→1, R→4, S→3
(D) P→2, Q→4, R→3, S→1

Q.69 An energy estimator has a per-shot standard deviation of 0.1 Ha. How many shots are needed for a standard error of 1.6 mHa (chemical accuracy)?

(A) about 1.6×10^5 **(B)** about 63 **(C)** about 3.9×10^3 **(D)** about 2.6×10^{-4}

Q.70 Under complete depolarisation the state becomes $\rho = I/2^n$. For the notebook's 4-qubit H_2 Hamiltonian the measured energy would then be

(A) −0.0971 Ha, the coefficient of `IIII`
(B) −1.1168 Ha
(C) −1.1373 Ha
(D) 0 Ha

Q.71 **Assertion (A):** With a finite number of shots, an estimated VQE energy can never fall below E_0.

Reason (R): For every density matrix ρ (pure or mixed, noisy or not), $\mathrm{tr}(\rho H) \geq E_0$.

(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A
(B) Both A and R are true, but R is *not* the correct explanation of A
(C) A is true, but R is false
(D) A is false, but R is true

Q.72 Zero-noise extrapolation: an energy measured at noise scale factors $\lambda = 1$ and $\lambda = 3$ gives -1.10 Ha and -1.00 Ha. A linear extrapolation to $\lambda = 0$ gives

(A) -1.10 Ha **(B)** -1.15 Ha **(C)** -1.05 Ha **(D)** -1.20 Ha

Q.73 `VQE.ipynb` is run with `BACKEND = "gpu"` on a machine where `cudaq.num_available_gpus()` returns 0. What happens?

(A) The notebook runs on the GPU emulator `tensornet` instead

(B) CUDA-Q silently falls back to `qpp-cpu`

(C) `resolve_backend` raises a `RuntimeError` explaining that no NVIDIA GPU is visible; no GPU target is selected

(D) The run starts and crashes later with an out-of-memory error

Q.74 **Statement I:** For the Ising chain, the notebook computes its reference energy by sparse diagonalisation for every N.

Statement II: For `"H2"` and `"custom"` Hamiltonians, an exact reference is computed only up to `REFERENCE_MAX_QUBITS = 18` qubits.

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true, but Statement II is false

(D) Statement I is false, but Statement II is true

Q.75 The notebook's exact Ising reference maps the chain to free fermions and computes $E_0 = -\frac{1}{2}\sum|\lambda(iA)|$ for a real antisymmetric matrix A. For $N = 100$ qubits, A has size

(A) 4×4 **(B)** 200×200 **(C)** 100×100 **(D)** $2^{100} \times 2^{100}$

Q.76 In the recorded run the quantum execution time was 69.10 s for 300 energy evaluations, while the notebook's pre-run estimate was ~ 102.9 s. The average time per evaluation, and the ratio estimate/measured cell time (70.1 s), are about

(A) 0.0032 s and 1.5 **(B)** 69 s and 1.5

(C) 0.23 s and 0.68 **(D)** 0.23 s and 1.5

Q.77 For LiH (STO-3G, 12 spin-orbitals under Jordan–Wigner), freezing the doubly occupied Li 1s core orbital removes how many qubits?

(A) 2, leaving 10 qubits **(B)** 0; frozen orbitals still need qubits

(C) 4, leaving 8 qubits **(D)** 1, leaving 11 qubits

Q.78 Generated with PySCF and OpenFermion (STO-3G, Jordan–Wigner, bond length 1.595 Å), the LiH qubit Hamiltonian has how many Pauli terms, compared with 15 for H_2?

(A) 4096 **(B)** 144 **(C)** 631 **(D)** 30

Paragraph for Questions 79 and 80

Previously recorded default run of ***VQE.ipynb*** (CUDA-Q 0.16.0, target `qpp-cpu`, 4 vCPU): Ising chain with $N = 15$, $J = 1$, $h = 0.5$ (29 Pauli terms); hardware-efficient ansatz with 2 layers (45 parameters, 73 gates); COBYLA stopped with "the objective function has been evaluated MAXFUN times" after 300 evaluations; $E_{\mathrm{VQE}} = -15.0533898284$, exact $E_0 = -15.0825179883$; quantum execution time 69.10 s; the optimised state gave $|000\ldots0\rangle$ with probability 0.7005. The product state $|000\ldots0\rangle$ itself has energy -14.

Q.79 Measured from the product state $|000\ldots0\rangle$ ($E = -14$), what fraction of the remaining energy gap to E_0 did VQE recover?

(A) 97.3 % **(B)** 99.8 % **(C)** 70.1 % **(D)** 2.7 %

Q.80 Which change is most likely to reduce the remaining error of this run?

(A) More final sampling shots

(B) Switching from the free-fermion reference to sparse diagonalisation

(C) More evaluations (the budget was exhausted) and, if needed, more ansatz layers

(D) Reducing `MAX_EVALUATIONS` to stop earlier

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